

Results - Lab8

```
1 BEGIN;
2
3 DROP TABLE IF EXISTS emp;
4 CREATE TABLE emp (
5     id integer PRIMARY KEY,
6     name varchar(30),
7     age integer,
8     dept_code integer,
9     office_loc varchar(100),
10    salary integer DEFAULT 0
11 );
12
13 INSERT INTO emp VALUES
14     (1, '张三', 22, 1, '宝安', 6000),
15     (2, '李四', 31, 1, '宝安', 7000),
16     (3, '王五', 25, 2, '福田', 6000),
17     (4, '赵六', 24, 1, '宝安', 5000),
18     (5, '庄七', 22, 3, '光明', 8000),
19     (6, '康八', 45, 2, '福田', 15000),
20     (7, '聂九', 34, 3, '光明', 7500),
21     (8, '刘二麻子', 56, 4, '光明', 17000),
22     (9, '孙小毛', 17, 1, '宝安', 3000),
23     (10, '陈老大', 37, 1, '宝安', 7000);
24
25 COMMIT;
26
27 SELECT count(*) AS tot FROM emp;
28
29 SELECT id, name, salary
30     FROM emp
31     ORDER BY id;
32
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://postgres:postgres@127.0.0.1:5432/lab8" -f './init.sql'
```

```
BEGIN
psql:init.sql:3: NOTICE:  table "emp" does not exist, skipping
DROP TABLE
CREATE TABLE
INSERT 0 10
COMMIT
tot
-----
 10
(1 row)
```

id	name	salary
1	张三	6000
2	李四	7000
3	王五	6000
4	赵六	5000
5	庄七	8000
6	康八	15000
7	聂九	7500
8	刘二麻子	17000
9	孙小毛	3000
10	陈老大	7000

```
(10 rows)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://postgres:postgres@127.0.0.1:5432/lab8" -f './test1.sql'
```

```
BEGIN
DELETE 0
UPDATE 1
id | name | salary
-----+-----+-----
 5 | 庄七 |  9000
(1 row)

COMMIT
id | name | salary
-----+-----+-----
 5 | 庄七 |  9000
(1 row)
```

```
[root@iZwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://postgres:postgres@127.0.0.1:5432/lab8" -f './test2.sql'
```

```
BEGIN
DELETE 1
UPDATE 1
id | name | salary
-----+-----+-----
 6 | 康八 | 14000
(1 row)

ROLLBACK
id | name | salary
-----+-----+-----
 1 | 张三 |  6000
 6 | 康八 | 15000
(2 rows)
```

```
[root@izwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://postgres:postgres@127.0.0.1:5432/lab8" -f './test3.sql'
```

```
SAVEPOINT
```

```
UPDATE 1
```

id	name	salary
5	庄七	10000
6	康八	14000
7	聂九	7500

(3 rows)

```
ROLLBACK
```

id	name	salary
5	庄七	10000
6	康八	15000

(2 rows)

```
UPDATE 1
```

```
COMMIT
```

id	name	salary
5	庄七	10000
6	康八	15000
7	聂九	6500

(3 rows)

```
[root@iZwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://po
'./test4.sql'
```

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000
6	康八	45	2	福田	15000
8	刘二麻子	56	4	光明	17000
9	孙小毛	17	1	宝安	3000
10	陈老大	37	1	宝安	7000
7	聂九	34	3	光明	6500
5	庄七	22	3	光明	10500

(10 rows)

```
BEGIN
```

```
UPDATE 1
```

```
SAVEPOINT
```

```
UPDATE 1
```

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000
8	刘二麻子	56	4	光明	17000
9	孙小毛	17	1	宝安	3000
10	陈老大	37	1	宝安	7000
7	聂九	34	3	光明	6500
5	庄七	22	3	光明	11500
6	康八	45	2	福田	14000

(10 rows)

ROLLBACK

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000
6	康八	45	2	福田	15000
8	刘二麻子	56	4	光明	17000
9	孙小毛	17	1	宝安	3000
10	陈老大	37	1	宝安	7000
7	聂九	34	3	光明	6500
5	庄七	22	3	光明	11500

(10 rows)

[root@iZwz989thjg5p4g6ibyphsZ Lab8]# psql "postgresql://postgres:postgres@localhost:5432/postgres" -f ./test5.sql

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000
6	康八	45	2	福田	15000
8	刘二麻子	56	4	光明	17000
9	孙小毛	17	1	宝安	3000
10	陈老大	37	1	宝安	7000
7	聂九	34	3	光明	6500
5	庄七	22	3	光明	10500

(10 rows)

BEGIN

TRUNCATE TABLE

COMMIT

id	name	age	dept_code	office_loc	salary
----	------	-----	-----------	------------	--------

(0 rows)

psql:test5.sql:9: WARNING: there is no transaction in progress

ROLLBACK

id	name	age	dept_code	office_loc	salary
----	------	-----	-----------	------------	--------

(0 rows)

[root@iZwz989thjg5p4g6ibyphsZ Lab8]#

psql (17.0)

Type "help" for help.

lab8_2=# BEGIN TRANSACTION;

BEGIN

lab8_2=*# INSERT INTO emp VALUES (1,'张三',22,1,'宝安',6000);

INSERT 0 1

lab8_2=*# SELECT * FROM emp;

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000

(1 row)

lab8_2=*# COMMIT;

COMMIT

lab8_2=# INSERT INTO emp VALUES (2,'李四',31,1,'宝安',7000);

INSERT 0 1

lab8_2=# COMMIT;

WARNING: there is no transaction in progress

COMMIT

lab8_2=# SELECT * FROM emp;

id	name	age	dept_code	office_loc	salary
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000

(2 rows)

lab8_2=#

```
lab8_2=# BEGIN TRANSACTION;  
BEGIN
```

```
lab8_2=# SELECT * FROM emp;
```

id	name	age	dept_code	office_loc	salary
-----+-----+-----+-----+-----+-----					

(0 rows)

```
lab8_2=# SELECT * FROM emp;
```

id	name	age	dept_code	office_loc	salary
-----+-----+-----+-----+-----+-----					
1	张三	22	1	宝安	6000

(1 row)

```
lab8_2=# SELECT * FROM emp;
```

id	name	age	dept_code	office_loc	salary
-----+-----+-----+-----+-----+-----					
1	张三	22	1	宝安	6000
2	李四	31	1	宝安	7000

(2 rows)

```
lab8_2=# select * from emp where id=5;
```

id	name	age	dept_code	office_loc	salary
-----+-----+-----+-----+-----+-----					

(0 rows)

Type 'help' for help.

```
lab8_3=# SET default_transaction_isolation='repeatable read';
SET
```

```
lab8_3=# BEGIN TRANSACTION;
BEGIN
```

```
lab8_3==# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
(0 rows)
```

```
lab8_3==# INSERT INTO emp VALUES (3,'王五',25,2,'福田',6000);
INSERT 0 1
```

```
lab8_3==# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
  3 | 王五 |  25 |          2 | 福田      |   6000
(1 row)
```

```
lab8_3==# COMMIT;
COMMIT
```

```
lab8_3=# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
  3 | 王五 |  25 |          2 | 福田      |   6000
(1 row)
```

```
lab8_3=#
```



```

lab8_3=# SET default_transaction_isolation='repeatable read';
SET
lab8_3=# BEGIN TRANSACTION;
BEGIN
lab8_3=*# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
(0 rows)

lab8_3=*# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
(0 rows)

lab8_3=*# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
(0 rows)

lab8_3=*# COMMIT;
COMMIT
lab8_3=# SELECT * FROM emp;
 id | name | age | dept_code | office_loc | salary
-----+-----+-----+-----+-----+-----
  3 | 王五 |  25 |          2 | 福田      |  6000
(1 row)

lab8_3=# 

```

```
lab8_3=# SET default_transaction_isolation='serializable';
SET
```

```
lab8_3=# BEGIN TRANSACTION;
BEGIN
```

```
lab8_3=*# SELECT * FROM emp
```

```
lab8_3-*# ;
```

id	name	age	dept_code	office_loc	salary
3	王五	25	2	福田	6000

(1 row)

```
lab8_3=*# INSERT INTO emp VALUES (4,'赵六',24,1,'宝安',5000);
INSERT 0 1
```

```
lab8_3=*# SELECT * FROM emp;
```

id	name	age	dept_code	office_loc	salary
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000

(2 rows)

```
lab8_3=*# COMMIT;
COMMIT
```

```
lab8_3=# SELECT * FROM emp;
```

id	name	age	dept_code	office_loc	salary
3	王五	25	2	福田	6000
4	赵六	24	1	宝安	5000

[root@12w2989cnj95p4g81bypnsz-Lab8]# psql postgresql://postgres:postgres@127

```
3 | 王五 | 25 |          2 | 福田          | 6000
(1 row)
```

lab8_3=# SELECT * FROM emp;

```
id | name | age | dept_code | office_loc | salary
----+-----+-----+-----+-----+-----
3 | 王五 | 25 |          2 | 福田          | 6000
(1 row)
```

lab8_3=# SELECT * FROM emp;

```
id | name | age | dept_code | office_loc | salary
----+-----+-----+-----+-----+-----
3 | 王五 | 25 |          2 | 福田          | 6000
(1 row)
```

lab8_3=# INSERT INTO emp VALUES (5,'庄七',22,3,'光明',8000);

ERROR: could not serialize access due to read/write dependencies among trans

DETAIL: Reason code: Canceled on identification as a pivot, during write.

HINT: The transaction might succeed if retried.

lab8_3=# SELECT * FROM emp;

ERROR: current transaction is aborted, commands ignored until end of transac

lab8_3=# COMMIT;

ROLLBACK

lab8_3=# SELECT * FROM emp;

```
id | name | age | dept_code | office_loc | salary
----+-----+-----+-----+-----+-----
3 | 王五 | 25 |          2 | 福田          | 6000
4 | 赵六 | 24 |          1 | 宝安          | 5000
(2 rows)
```